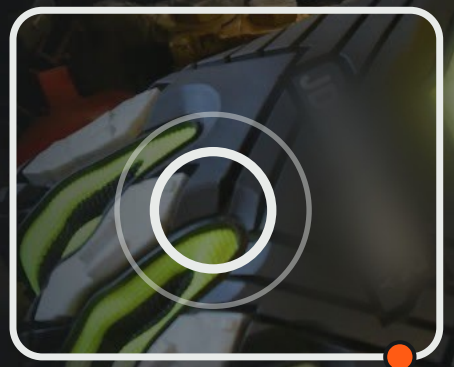
praxis

STEP 4/9

Instructions that can finally see how you're doing

Software for smart glasses that coaches new maintenance technicians from a company's existing manuals.



Seat the clamp past the hose bead

Startups @ Harvard · October 20, 2026 · Monish Saravana

Problem

Training new technicians is expensive

Each new hire learns by shadowing an experienced technician, and that time comes off the job.

Total cost to hire and ramp up one maintenance technician

$10–15k

including recruiting fees, onboarding time and lost productivity

---

~255,000

US maintenance and repair openings a year, each one needing someone to train them

In a 2017 survey of 35 aviation repair stations, new hires took 9–14 months to become profitable.

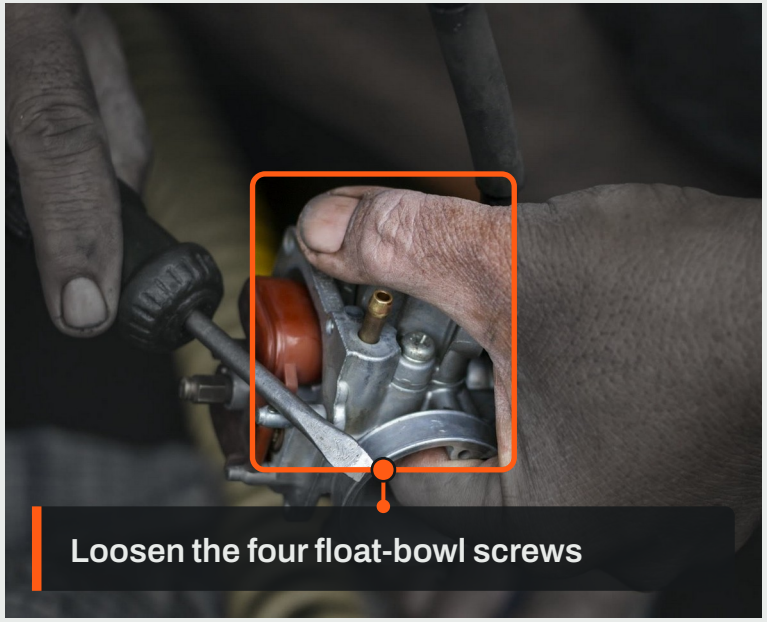
> **“The hardest AND most expensive part is training new techs …”**
> Casey Czarnowski, Healthcare Sterile Processing Association (HSPA), discussed this week



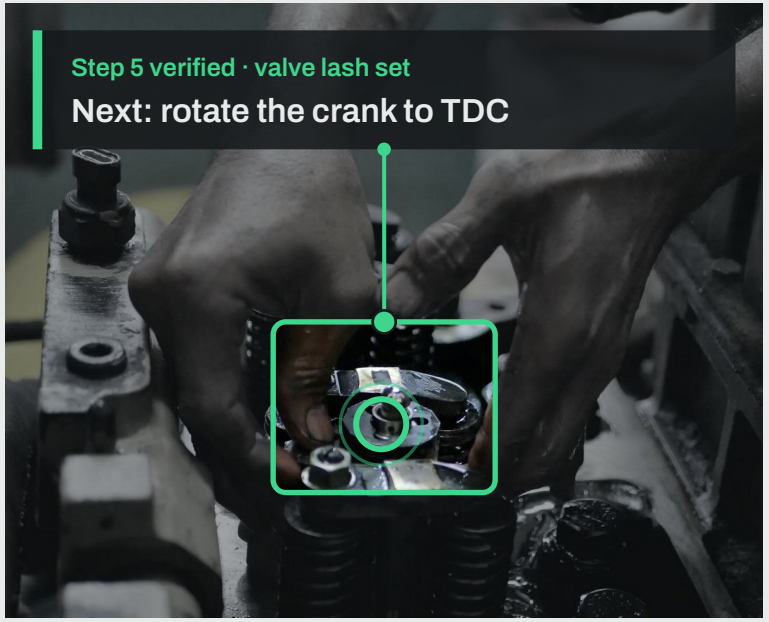
Praxis coaches the job through smart glasses

1 Upload the manual → 2 Praxis compiles it → 3 Tech wears glasses → 4 Praxis tracks the job → 5 Verifies key steps

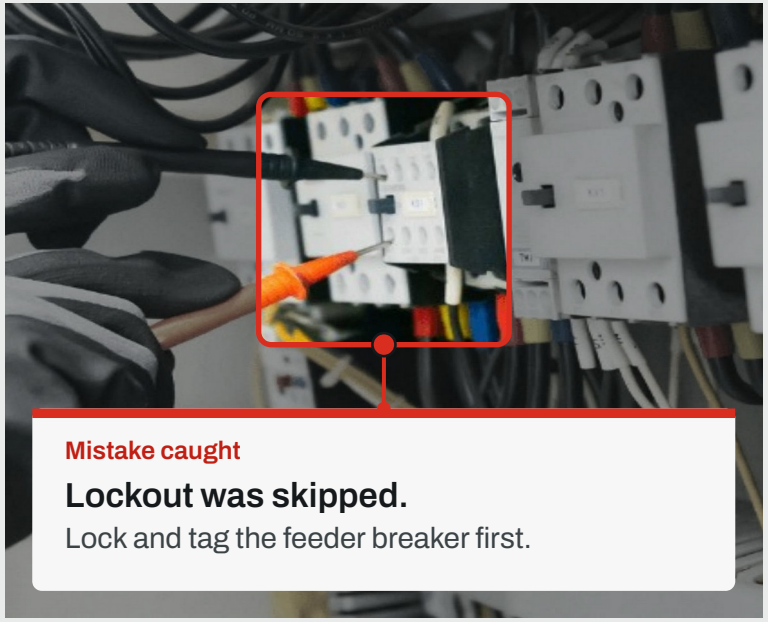
1 Says the next step



2 Checks the work



3 Catches a mistake



Built for new technicians in plant maintenance, facilities, HVAC service and equipment repair.

This is already a working software system

1 A manager uploads the manuals they already have

3 DOCUMENTS

DOCUMENT	PAGES	SECTIONS	LIKELY PROCEDURES
FIST 1-1 · Hazardous Energy Control Program FIST_1-1.pdf · v1	97	131	4
FIST 4-1A · Maintenance Scheduling for Mechanical Equipment FIST_4-1A_2020.pdf · v1	222	234	3
FIST 3-8 · Protective Relays: Operation, Maintenance and Field Tests FIST_3-8_2021.pdf · v1	125	97	0

2 Praxis compiles a procedure; the manager reviews and approves it

High Voltage Board Repair (Electric Screwdriver)

v1 · approved · live ▾ · Approved · live

Source **High-voltage board repair (electric screwdriver)** v1 · High Voltage Board Repair Procedure (Electric Screwdriver) · Steps from split call · Sent 6,569 chars of a 1-page document
Local tracking: document only · mark steps on a recording · Cost $0.26 · claude-opus-5-5 · Oct 18

WHOLE PROCEDURE

- **high voltage** HIGH VOLTAGE BOARD REPAIR PROCEDURE (ELECT... ×
- **high voltage** Observe the high voltage board for a moment ×
- **high voltage** Take the high voltage board and places it on the work area ×
- **electrical: 5 Volts** Adjust the power supply's voltage knob to set it to 5 Volts ×
- **disconnect the power** Disconnect the power supply cables ×
- **temperature: 200 degrees** Set the soldering iron temperature to 200 degrees using th... ×
- **temperature: 160 degrees** Reduce the soldering iron temperature to the minimum (1... ×
- **temperature: 160 degrees** Lower the soldering iron temperature to the minimum (160... ×
- **electrical: 5 Volts** Set the voltage knob of the power supply to 5 Volts ×

STEPS 20 · 0 safety-critical — Expand all

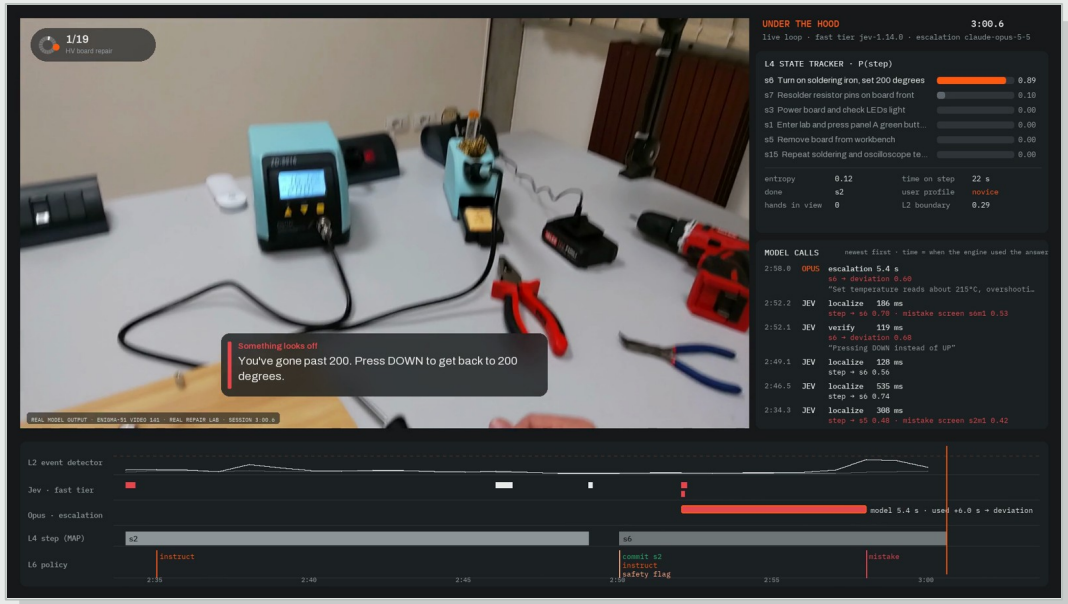
1 **Start recording and toggle bench lamp** — Sit at the workbench and say "Record" clearly, then wait for the b... — Safety wording in source — 1 card ›

What the compiler saw — 6,569 chars

Page 1 ↗

HIGH VOLTAGE BOARD REPAIR PROCEDURE (ELECTRIC SCREWDRIVER) Sit at the workbench Pronounc the voice command "Record" to start the recording; you will hear an audible signal for confirmation Turn on and off the lamp on the workbench while looking at it Leave the laboratory Enter the laboratory and closes the door Head towards Panel A and observes it for a moment Press the two green buttons on Panel A Move back to the workbench and sits down Observe the high voltage board for a moment Take the high voltage board and places it on the work area Connect the battery to the electric screwdriver Remove the screws from the workbench using the electric screwdriver Secure the board to the workbench using the electric screwdriver Observe the power supply Turn on the power supply using the socket switch Adjust the power supply's current knob until the green LED lights up Adjust the power supply's voltage knob to set it to 5 Volts Connect the power supply cables to the board's power points Watch the board for a few seconds to check if the red and green LEDs turn on Turn off the board's power supply using the socket switch and wait for the board's LEDs to turn off Set the current and voltage knobs of the power supply back

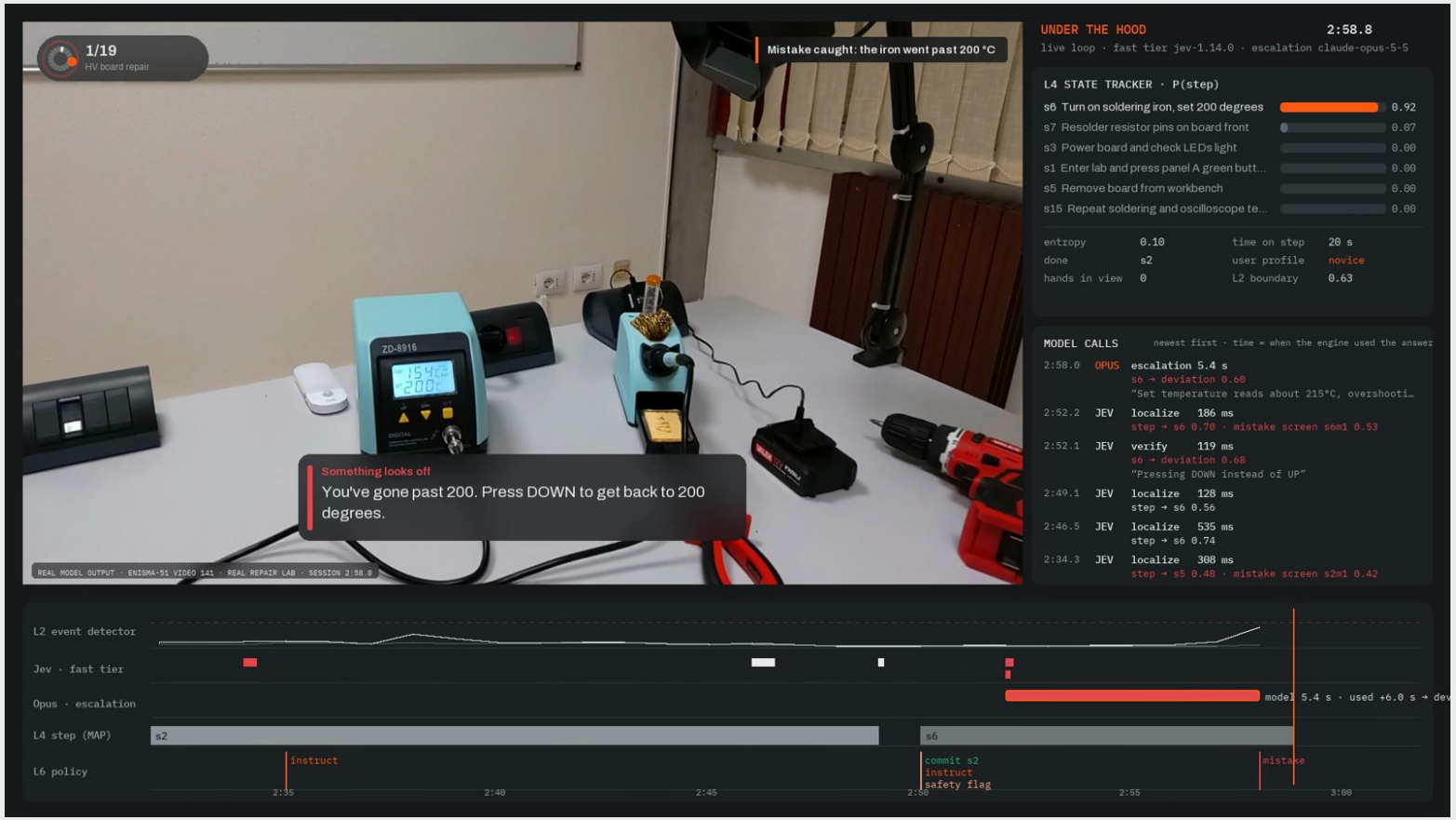
3 The live app tracks the job (engineer view)



- **Company backend: upload, index, compile, approve** built
- **Tracking engine and evaluation harness** built
- **Live app with spoken coaching** built, real time on video
- **Meta glasses adapter** coded, not device-tested
- **Viture/NVIDIA glasses adapter** coded, not device-tested
- **Android XR glasses** planned

Live test

We already run a real repair procedure live



Real repair, start to finish

16:40

a high-voltage board repair, filmed first-person in an industrial lab

Training on this exact job

None

Praxis started from the lab's written procedure

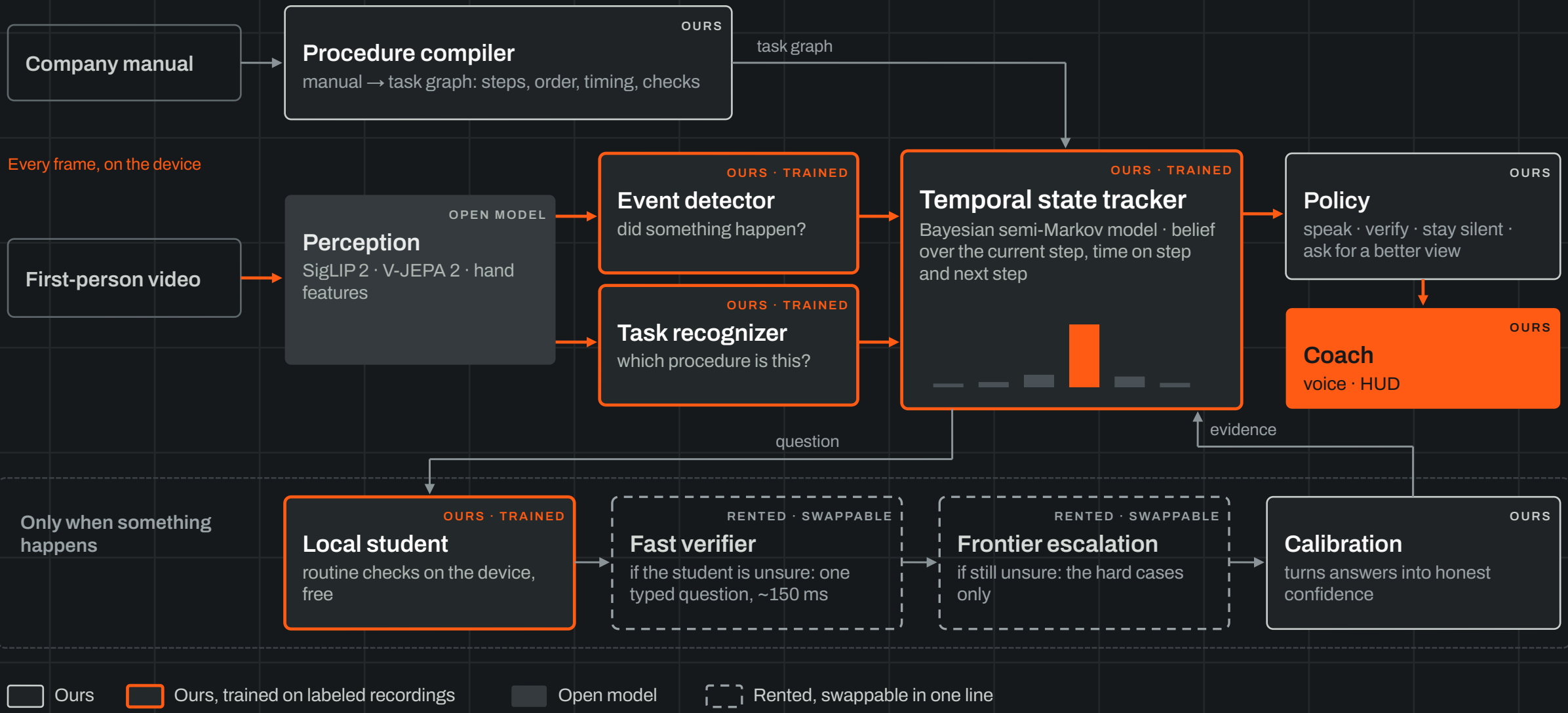
Model cost

≈$1

of model calls per coached hour, measured across our evaluation

How it works

The system tracks the job instead of querying a model constantly

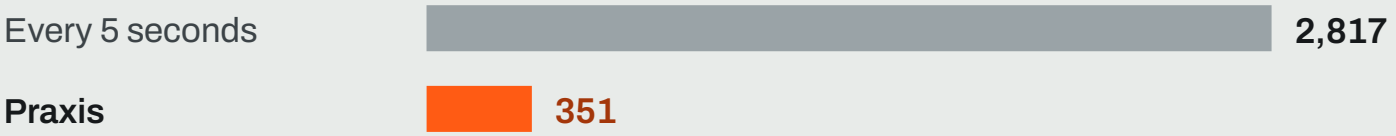


Measured

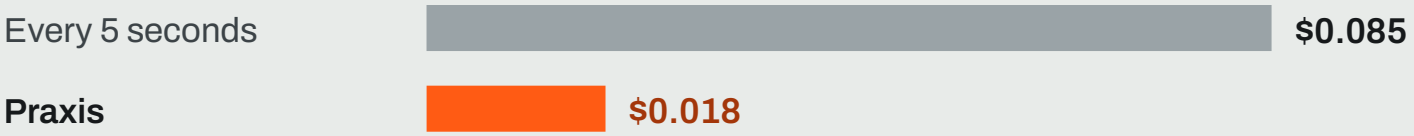
Same frontier model, better system

We compared Praxis with calling the same model (Claude Opus 5.5) every 5 seconds, on 48 first-person videos neither had seen.

8× fewer calls to the frontier model



5× cheaper per minute of video



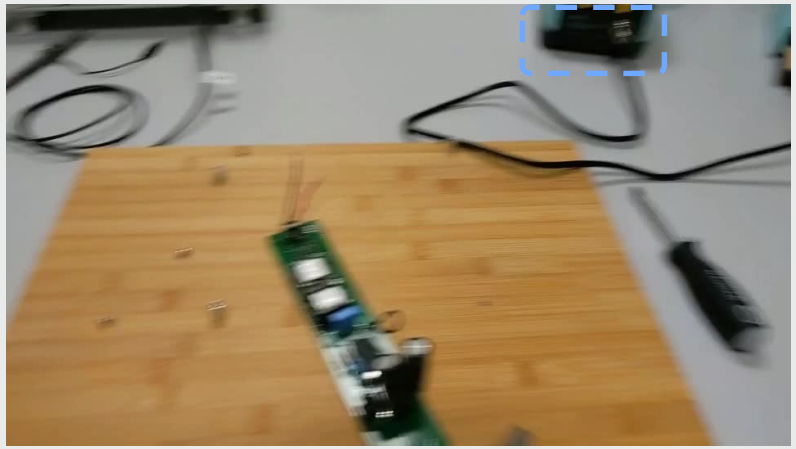
About $1

of model calls per coached hour, against $200 a month per trainee

Better tracking

It follows the step more accurately than the baseline while asking the frontier model 8× less

When it can't see, it says so



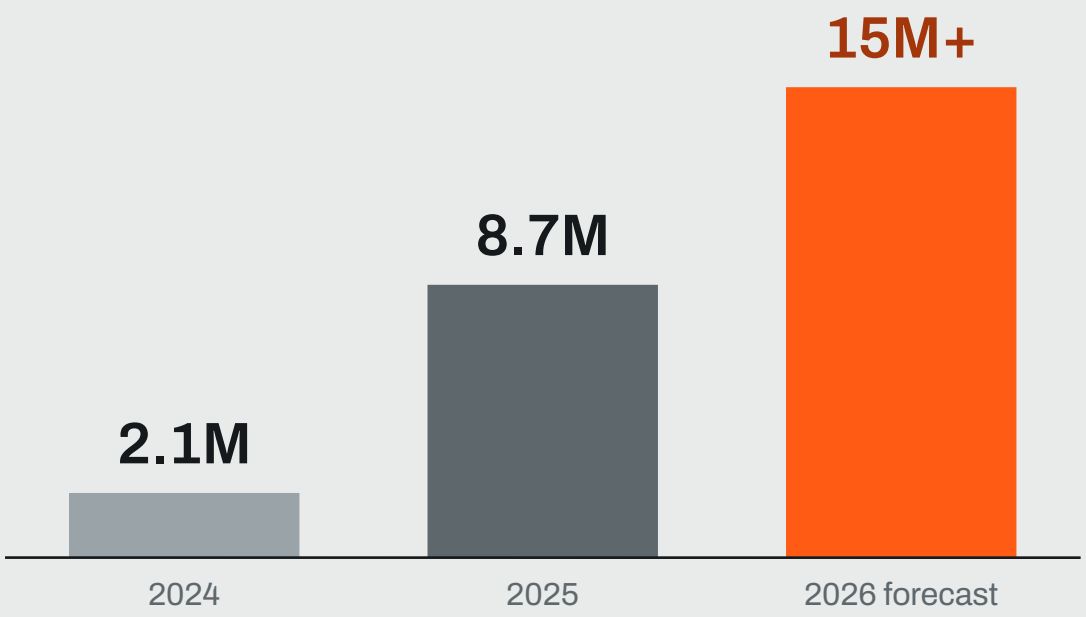
> **“Look closely at the soldering station display, upper right.”**
>
> Praxis at 4:25 in the live repair, with the display half out of frame

77–98% of blurred or blocked checks: Praxis asked to see it again

Up to 73% the every-5-seconds baseline named the wrong step with confidence

Smart-glasses hardware is finally open to software

AI glasses shipped per year, worldwide

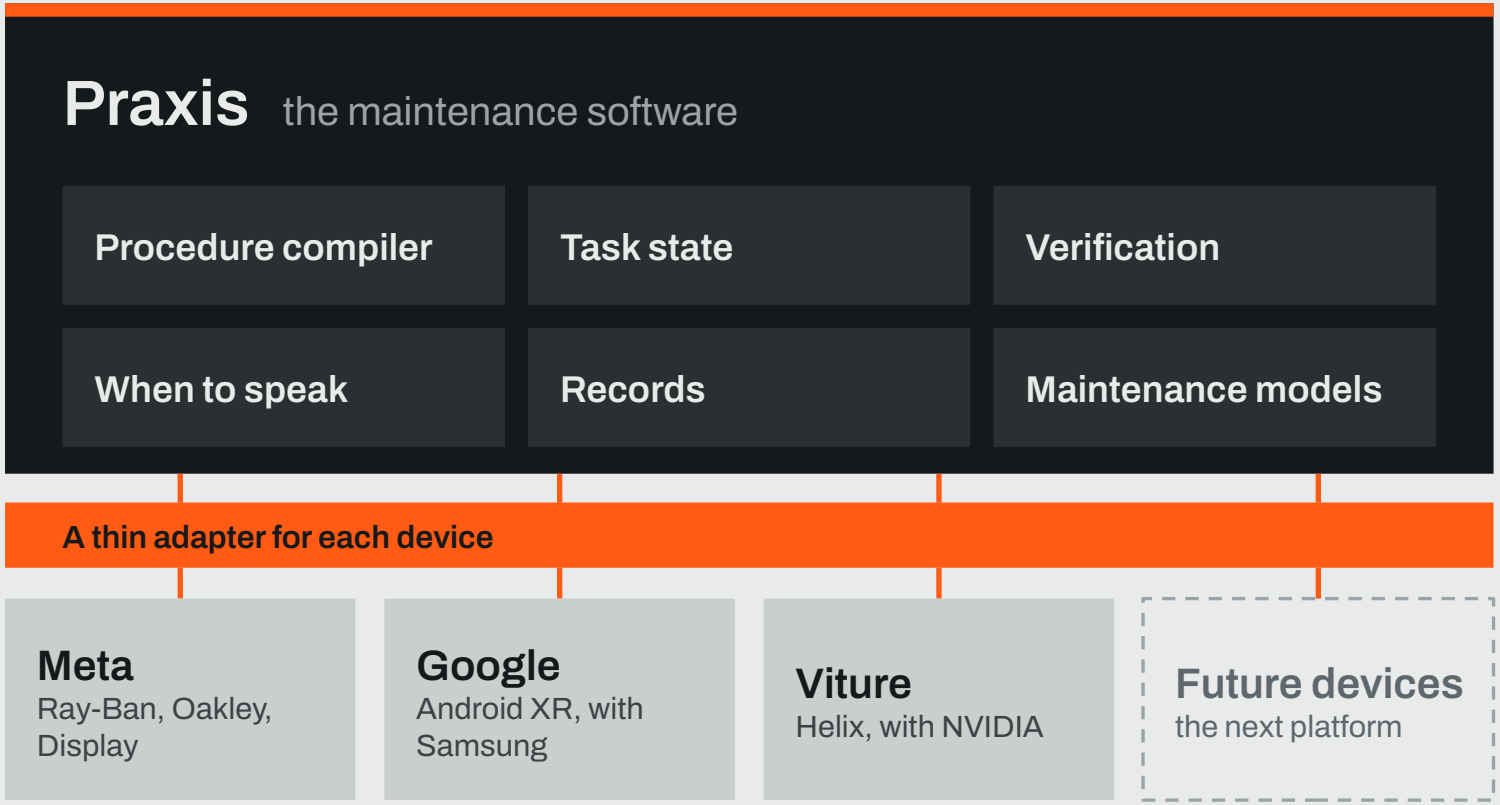


Platforms opening to outside developers

2025	Meta opens a developer preview: outside apps can use its glasses' camera
Sept 2026	Meta's toolkit reaches 1.0: camera, microphones and speakers for outside apps
Fall 2026	Google's first Android XR glasses arrive: audio, with a camera
Q1 2027	Viture and NVIDIA ship Helix: safety glasses that coach workers on procedures in real time

Three years ago we would have had to build the glasses.
Now we build the software that runs on them.

We don't need to win the glasses hardware race



They build the glasses. Every new device is one more adapter for Praxis.

Other approaches

Procedure software MaintainX, Tulip, Augmentir
Steps on a screen. It can't see the work.

Fixed cameras Retrocausal, Invisible AI
Watch one station. They can't follow a tech around a machine.

General assistants Meta AI, Gemini Live
Answer about what's in view, with no procedure or step state.

Viture Helix + NVIDIA Our closest competitor
Real-time coaching sold with its own glasses. We want Praxis to run on Helix too.

Maintenance is where we start, and Praxis pays for itself

Maintenance new-hire onboarding in the US, per year

~255k new maintenance techs × **$200** a month per trainee × **~6** months of onboarding ≈ **$306M** a year, at our pricing hypothesis

- **Start** maintenance new-hire onboarding
- **Next** all maintenance and field service
- **Then** technical onboarding in manufacturing, utilities, aviation, telecom, labs

One new technician

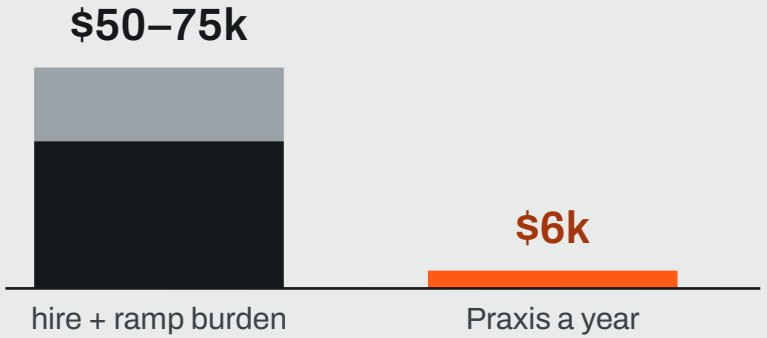
Hire + ramp burden today	**$10–15k**
Praxis, six months	**$1,200**
Our cost to serve	**~$270**

~77% gross margin today, ~90% with our local model

Pays for itself if it removes roughly 8–12% of that burden.

A 50-technician maintenance team

about 5 new hires a year



- **We start with small and mid-size maintenance teams.** 98% of US manufacturers have fewer than 500 employees, and there one maintenance manager can approve a paid pilot. The pilot measures senior-tech hours, days until a new tech works alone, and rework.

We sell to maintenance managers, one paid pilot at a time

1 **Maintenance manager**

at a plant, facilities team, HVAC company or repair shop

2 **Demo on their own manual**

a procedure their new techs already struggle with

3 **Short paid pilot**

the success metric is agreed first, like days until a new tech works alone

4 **Annual agreement**

$200 a month per active-trainee license

5 **Expand**

every new hire, then other sites

How pricing works

One active-trainee license = one employee being onboarded: $200 a month, unlimited coaching sessions

Five techs onboarding at once = five licenses. When one finishes, the license moves to the next hire

Glasses are shared site equipment. 10 licenses with 3 people coached at the same time needs about 3 pairs

We don't resell hardware. Sites buy supported glasses directly

Second path: trade schools

Students learn hands-on skills with Praxis and bring it to their first employer.

> **“We would be open to an experiment like this.”**
>
> Sonny Hunt, COO, The Fiber School (FiberOptic.com), discussed this week

Founder

I've built real-time vision systems for the physical world, and I've helped grow startups

Engineering and research

60,000+
video frames

LipTrac
Real-time visual speech recognition. Conv3D + bidirectional GRU, tested with 12 participants. Provisional patent filed.

217.9%
better collection than conventional methods

OCTOPAS
Autonomous oil-spill cleanup: aerial computer vision, path planning and marine robotics. Published; provisional patent filed.

7 years
FTC team captain

Competitive robotics
World Championship qualifier. Autonomous control, sensor fusion, OpenCV and path planning. My open-source FTC Coach has 25,000+ users.

Startups and growth

$50k
MRR, from under $1k

Qurios Ventures · co-founder, CMO
Built a 14,000+ member community in about four months. Acquired by Crimson Education for $1.3M.

13
person growth team

Claisen Health · co-founder, CMO
Ran acquisition across UGC, influencers and Reddit as the company raised $200k from angels.

$300k
attributable revenue

Acely AI (YC W18)
One of 6 marketing interns picked from 1,500 applicants.

Harvard College Applied Mathematics + Computer Science

Coaching comes first. Our own models and skill records come next

Now

Maintenance coaching and verification

Paid pilots. New techs coached from their own company's manuals.

Next

Maintenance-specific models

With customer permission, real maintenance recordings make our state and checking models better and cheaper.

Then

Verified skills and certification partners

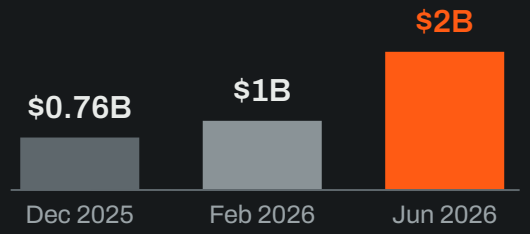
Techs and employers build verified skill histories in Praxis. Certified techs bring Praxis to their next job.

Later

Physical AI, two ways in

- **A Build our own robot models** trained on step-verified recordings customers opted in
- **B Sell to physical-AI labs** license consented data and our step checker

What AI labs pay for human training data keeps climbing



Mercor's run-rate from selling expert human data to AI labs

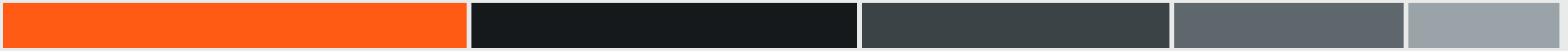
>$100M
a year that robotics companies already spend buying real-world data

>$1B
that Figure has committed to data and compute over the next 12 months

$3.2B → $16.3B
the AI training-data market, 2025 to the 2033 forecast

Ask

$10k gets us to the first paid pilot



$3,000

Model training

GPU time to retrain the step tracker, event detector and local checker on real maintenance footage

$2,500

Devices + capture

Supported glasses and capture gear to run Praxis end to end at a real site

$2,000

Maintenance data

Pay techs and trade-school students to record and label ~80 hours of real work, with consent

$1,500

Pilot inference

About 1,400 coached hours at today's measured cost

$1,000

Company + legal

Formation and the contracts a paid pilot needs

By April

Models retrained on real maintenance footage

Praxis running end to end on real glasses at a real site

A first paid pilot, with the success metric agreed before launch

For questions

Appendix

A1 Benchmark results and method

A2 Dataset, splits and denominators

A3 Learning curves

A4 Architecture with numbers

A5 Live session, engineer view

A6 Company backend

A7 What we train and what we rent

A8 Competitors in detail

A9 Market sizing math

A10 Unit economics and pricing risk

A11 Glasses hardware and adapters

A12 Data rights and privacy

A13 Safety

A14 Risks

A15 Why we don't start with consumers

A16 Founder background

A17 Revenue targets, certifications, robotics

Appendix A1

Benchmark results and method

SYSTEM (SAME 48 HELD-OUT CLIPS, TASK GIVEN)	STEP TRACKING	FRONTIER (OPUS 5.5) CALLS	COST / VIDEO-MIN
Baseline: Claude Opus 5.5 every 5 s, no task state	48%	2,817	$0.085
Praxis: fast tier (Jev) + Opus 5.5 escalation	54%	351 (+2,679 fast-tier)	$0.018
Praxis, fast tier only (no escalation)	53%	0	$0.001
Praxis + our local student on routine checks	52%	not broken out	$0.008
Our local layers only, no model calls	41% (49% retrained)	0	$0

Data. 48 held-out first-person clips (235 min) from CaptainCook4D, a public procedure dataset (cooking): 8 procedure types, 302 annotated steps, 170 annotated mistakes. Labels are the dataset's. Train and test never overlap.

Metric. Step tracking = share of frames where the tracked step matches the label. One run per system; no confidence intervals yet.

Mistakes (strict, judge-verified). Praxis alerts named a real annotated error 4 of 17 times; the baseline 2 of 3. Both catch a small share of the 170.

Degraded views (60 checks per condition). Praxis asked for a better view on 77–98% and was never confidently wrong; the baseline was wrong at ≥50% confidence on 12–73%.

Repair Q&A (ENIGMA-51). 11 of 15 technician questions answered correctly from the uploaded procedures, 2 partly, 2 wrong, blind-judged.

Dataset, splits and denominators

Train and test never overlap. Everything learned uses only the dataset's official train-split recordings

Small on purpose. A weekend-scale proof: the step-change detector trained on 24 recordings, the step models on 3 per procedure, the student on checks from 30

Tested on 48 held-out clips: 235 min, 302 annotated steps, 170 annotated mistakes, plus procedure types never seen in training

Design choices were made on train-split recordings only, never on the test clips

Labels are the dataset's, not ours. One run per system, no confidence intervals yet

LEARNING CURVES

Step models: 30.7% with 1 recording → 43.4% with 3 → 50.7% with 9 (the 3 procedures with 9)

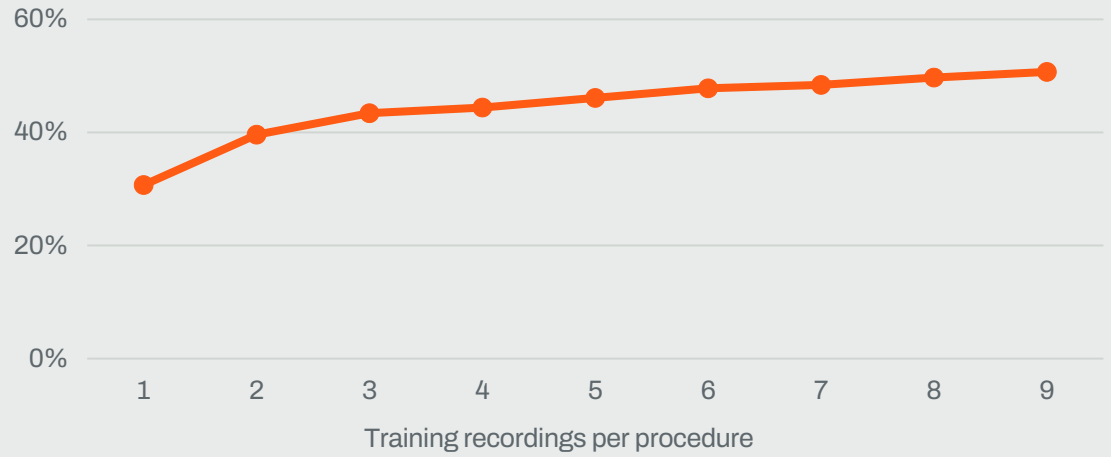
Student: 62.6% with 4 recordings → 72.5% with 30; the frontier teacher is 79.7%

Detector: noisy, no clear trend yet. Not claimed

Learning curves: step models and the local student

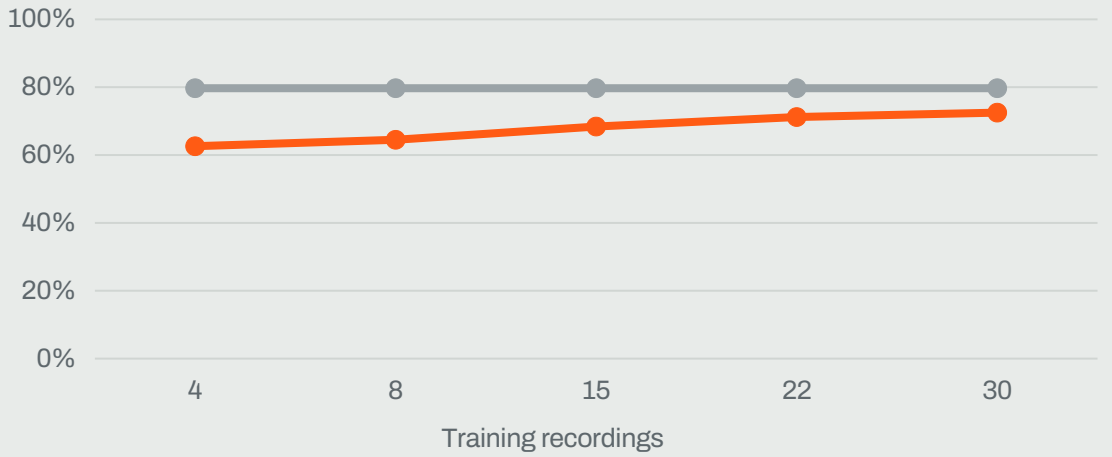
30.7% → 50.7%

recognizing the current step, on videos it never trained on



62.6% → 72.5%

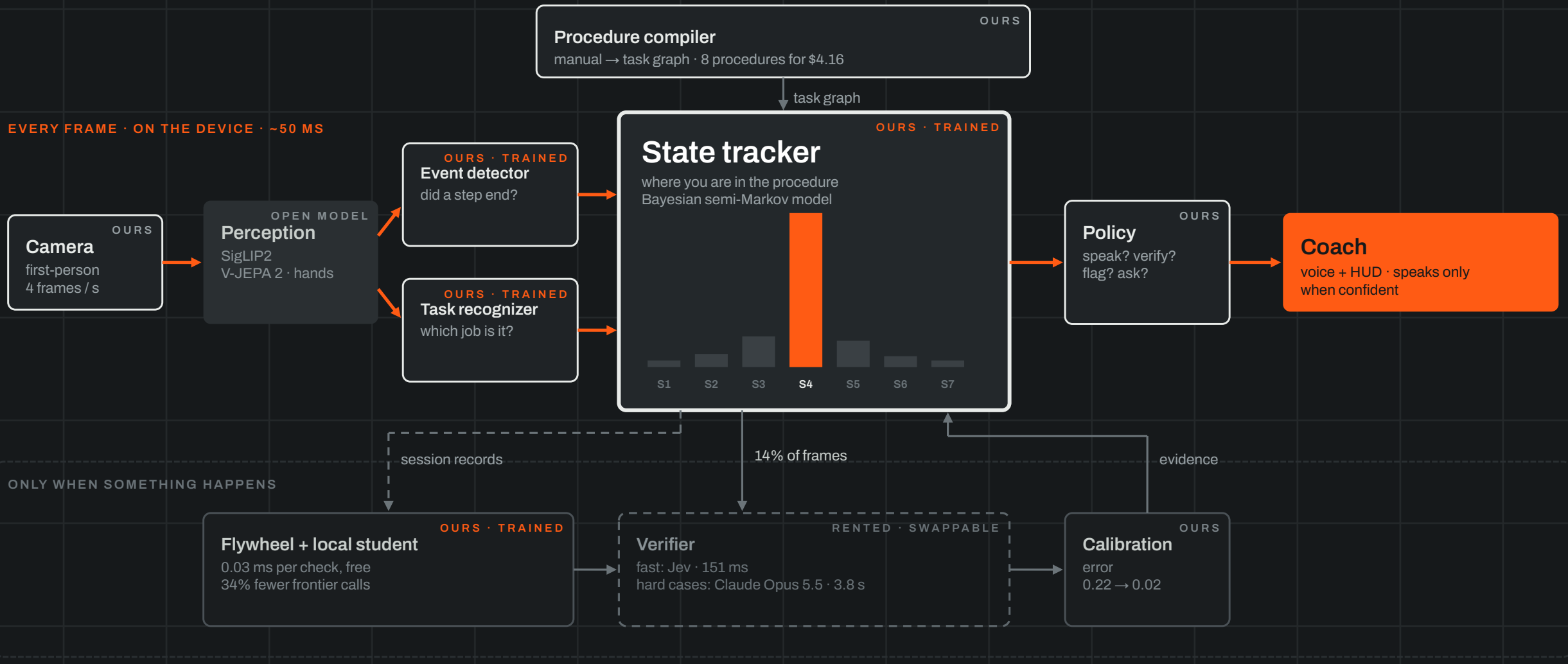
checking a step is done · grey: the frontier model, 79.7%



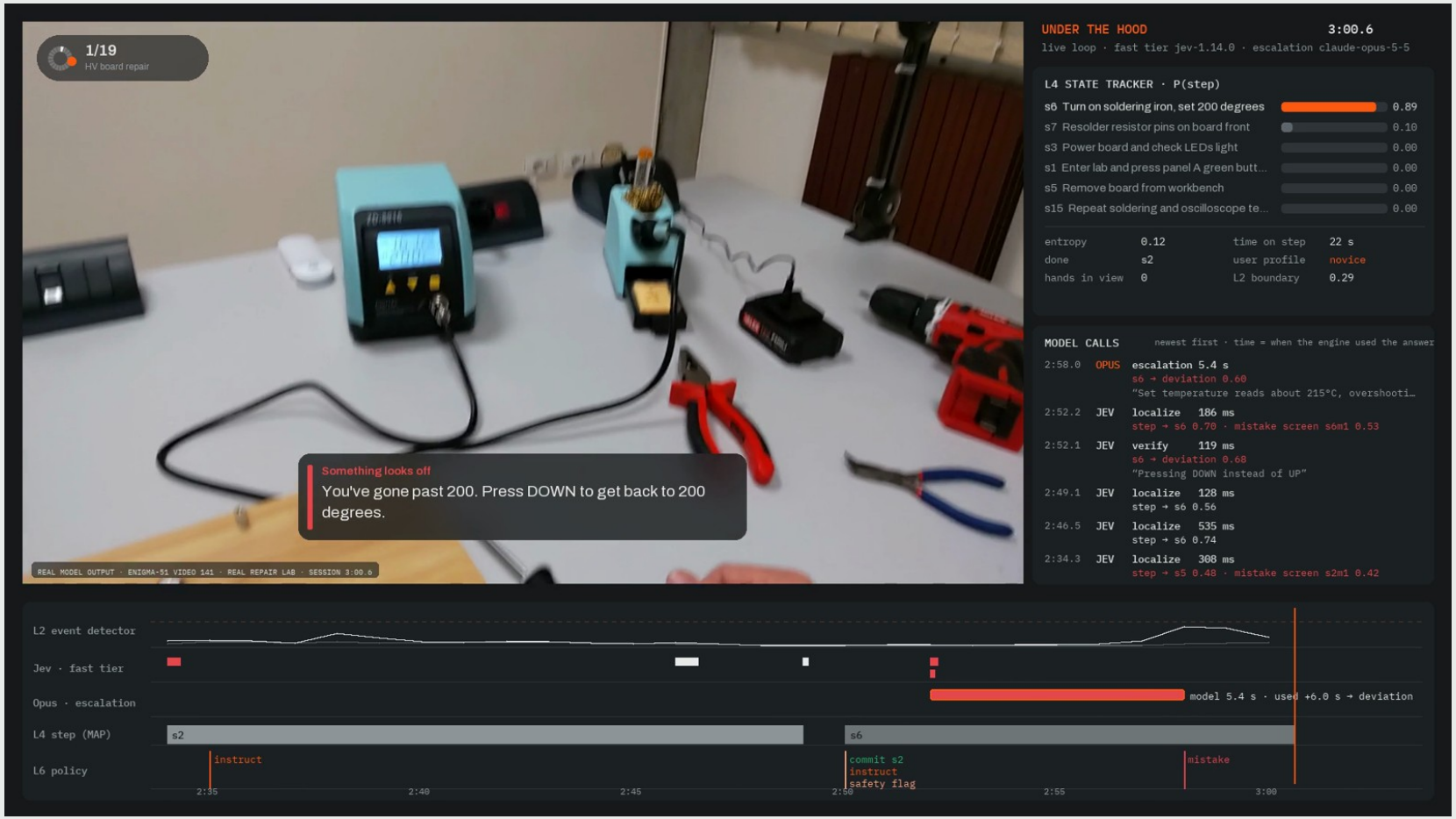
TESTED ONLY ON RECORDINGS IT NEVER SAW · BOTH STILL RISING AT THE LARGEST SIZE WE HAVE

Appendix A4

Full architecture. Orange runs on every frame; the bottom row only when something happens.



Live session, engineer view



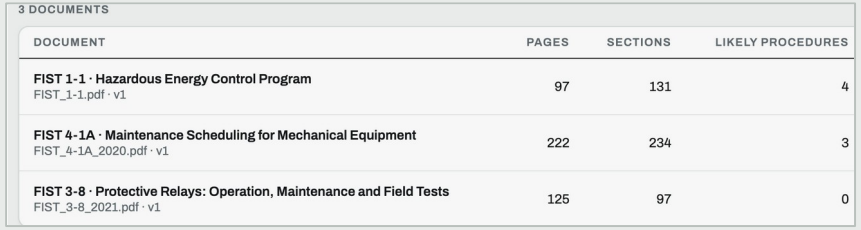
State tracker. Its belief over the 19 steps, the time on the current step, and whether the tech looks like a novice.

Model calls. Jev answers in 150 ms (median, 130 calls). Opus 5.5 double-checks 14 moments, about 4 s each.

Timeline. The event detector, both model tiers, the tracked step and every decision to speak or stay quiet.

Cost. $0.17 for this whole 16:40 run (about $0.60 an hour), with 0 errors. Across the evaluation, about $1 per coached hour.

Company backend: upload, compile, approve



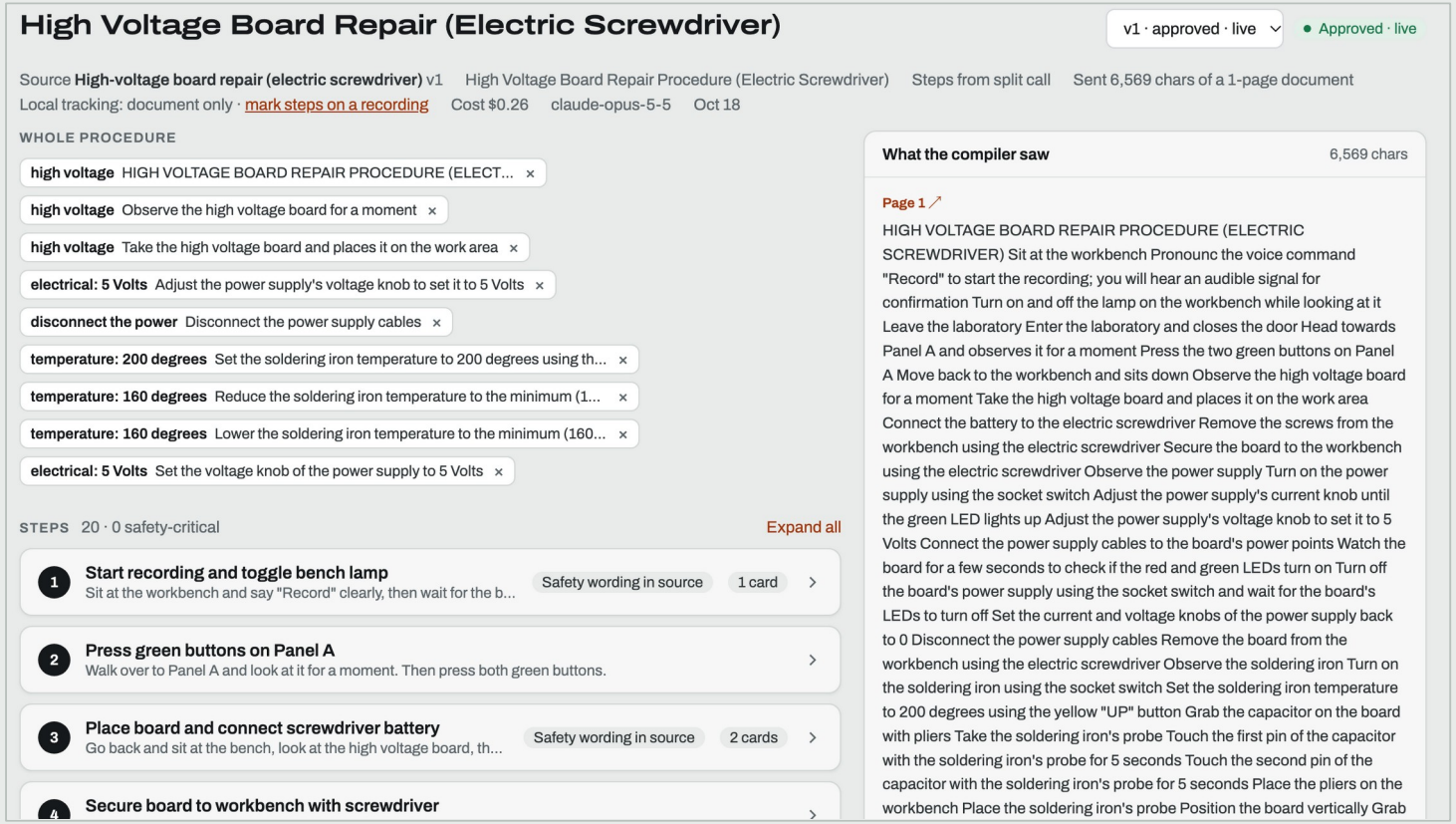
3 DOCUMENTS

DOCUMENT	PAGES	SECTIONS	LIKELY PROCEDURES
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FIST 4-1A · Maintenance Scheduling for Mechanical Equipment FIST_4-1A_2020.pdf · v1	222	234	3
FIST 3-8 · Protective Relays: Operation, Maintenance and Field Tests FIST_3-8_2021.pdf · v1	125	97	0

1. **Upload the manuals**
   A 97-page safety manual is indexed in 0.34 s on a laptop, for $0. Its procedures are found automatically.

2. **Review the coached steps**
   Each procedure compiles for about $0.19, with cards quoted from the manual. A manager approves it first.

3. **Coach through smart glasses**
   Live tracking and voice control. Meta and Viture adapters are built, not yet tested on a device.



High Voltage Board Repair (Electric Screwdriver)

v1 · approved · live · Approved · live

Source **High-voltage board repair (electric screwdriver)** v1 · High Voltage Board Repair Procedure (Electric Screwdriver) · Steps from split call · Sent 6,569 chars of a 1-page document
Local tracking: document only · mark steps on a recording · Cost $0.26 · claude-opus-5-5 · Oct 18

WHOLE PROCEDURE

- **high voltage** HIGH VOLTAGE BOARD REPAIR PROCEDURE (ELECT... ×
- **high voltage** Observe the high voltage board for a moment ×
- **high voltage** Take the high voltage board and places it on the work area ×
- **electrical: 5 Volts** Adjust the power supply's voltage knob to set it to 5 Volts ×
- **disconnect the power** Disconnect the power supply cables ×
- **temperature: 200 degrees** Set the soldering iron temperature to 200 degrees using th... ×
- **temperature: 160 degrees** Reduce the soldering iron temperature to the minimum (1... ×
- **temperature: 160 degrees** Lower the soldering iron temperature to the minimum (160... ×
- **electrical: 5 Volts** Set the voltage knob of the power supply to 5 Volts ×

STEPS 20 · 0 safety-critical — Expand all

1. **Start recording and toggle bench lamp** — Sit at the workbench and say "Record" clearly, then wait for the b... — Safety wording in source — 1 card
2. **Press green buttons on Panel A** — Walk over to Panel A and look at it for a moment. Then press both green buttons.
3. **Place board and connect screwdriver battery** — Go back and sit at the bench, look at the high voltage board, th... — Safety wording in source — 2 cards
4. **Secure board to workbench with screwdriver**

What the compiler saw — 6,569 chars

Page 1 ↗

HIGH VOLTAGE BOARD REPAIR PROCEDURE (ELECTRIC SCREWDRIVER) Sit at the workbench Pronounc the voice command "Record" to start the recording; you will hear an audible signal for confirmation Turn on and off the lamp on the workbench while looking at it Leave the laboratory Enter the laboratory and closes the door Head towards Panel A and observes it for a moment Press the two green buttons on Panel A Move back to the workbench and sits down Observe the high voltage board for a moment Take the high voltage board and places it on the work area Connect the battery to the electric screwdriver Remove the screws from the workbench using the electric screwdriver Secure the board to the workbench using the electric screwdriver Observe the power supply Turn on the power supply using the socket switch Adjust the power supply's current knob until the green LED lights up Adjust the power supply's voltage knob to set it to 5 Volts Connect the power supply cables to the board's power points Watch the board for a few seconds to check if the red and green LEDs turn on Turn off the board's power supply using the socket switch and wait for the board's LEDs to turn off Set the current and voltage knobs of the power supply back to 0 Disconnect the power supply cables Remove the board from the workbench using the electric screwdriver Observe the soldering iron Turn on the soldering iron using the socket switch Set the soldering iron temperature to 200 degrees using the yellow "UP" button Grab the capacitor on the board with pliers Take the soldering iron's probe Touch the first pin of the capacitor with the soldering iron's probe for 5 seconds Touch the second pin of the capacitor with the soldering iron's probe for 5 seconds Place the pliers on the workbench Place the soldering iron's probe Position the board vertically Grab

What we train and what we rent

PART	WHO	WHAT IT DOES
Event detector	OURS · TRAINED	A small temporal network that spots the moment a step ends
Task recognizer	OURS · TRAINED	Works out which procedure the tech is doing
State tracker	OURS · TRAINED	A Bayesian model of where the tech is in the procedure, with step models per procedure
Local checker	OURS · TRAINED	Learned from the frontier model; answers routine "is this step done?" checks for free
Perception	OPEN MODELS	SigLIP2, V-JEPA 2 and hand tracking turn each frame into features. Not trained by us
Verifier	RENTED	A frontier model answers hard questions on 14% of frames. Swappable in one line

WHY A BIG LAB WON'T

They sell one general model. Ours wraps any model, so their best one becomes our swappable verifier.

The hard part is data they don't have: first-person video of skilled work, lined up to the procedure, with verified steps.

And a business they don't run: each shop's manuals and records, and a pilot on site.

Appendix A8

Competitors in detail

APPROACH	EXAMPLES	WHAT IT DOES	HOW WE DIFFER
Shadowing	A senior tech	The real teacher today	About $44 an hour of senior time, one new tech at a time
Manuals + videos	PDFs, binders, YouTube	Explain what should happen	Can't see what the tech is doing
Procedure software	MaintainX, Tulip, Augmentir, Dozuki	Steps on a screen, sign-offs, photos; some compile instructions	A checkbox isn't proof. Natural integration partners
Fixed cameras	Retrocausal, Invisible AI, Tulip Vision	Watch one station; record, label and train per station	Ahead of us on live mistake detection at a station; can't follow a tech around a machine
General assistants	Meta AI, Gemini Live	Free help about what's in view	No procedure, step state, calibration or records
First-person, high-touch	Strivr	Per-customer capture and model (vendor docs); early access	No published accuracy; we start from the manuals a shop has
First-person, hardware-led	Viture Helix + NVIDIA XR AI	Real-time SOP coaching on its own safety glasses; ships Q1 2027, from $600	Closest. How SOPs get in and accuracy aren't published. We'd rather run on it
Toolkits	NVIDIA XR AI, Meta's toolkit, Android XR	Plumbing for apps	Complements: what we build on

Appendix A9

Market sizing math

	WHO (BLS)	MATH	PER YEAR
TAM	Installation, maintenance + repair 6,086,190 · electricians 757,220 · production 8,565,760 = 15.4M	15.4M × $50 a month × 12	≈ $9.2B
SAM	General maintenance + repair, machinery mechanics + millwrights, HVAC = 2.47M jobs, ~255k openings a year	255k × $200 × 6 months = $306M; 2.47M × $50 × 12 = $1.48B	≈ $1.8B
SOM (year 3)	1.5% of new maintenance techs ≈ 3,750 new techs a year, ~150 mid-size teams	3,750 × $200 × 6 months	≈ $4.5M ARR

Prices are hypotheses. $200 a month for coaching is a hypothesis; Tulip lists ~$100/month (Essentials) and ~$250/month (Professional) per interface, which anchors it but doesn't prove willingness to pay. $50 a month for every-worker records is a guess

Openings include people changing jobs, so new-tech counts are an upper bound

Not counted: other countries, field service outside these groups, aviation maintenance, and the robotics upside

Appendix A10

Unit economics and pricing risk

DESIGN	$ / VIDEO-MIN	$ / TRAINEE-MONTH, 2 H/DAY	MARGIN AT $200
Frontier model every 5 s	$0.085	about $214	negative
Praxis engine today	$0.018	about $45	about 77%
Praxis + our local model	$0.008	about $20	about 90%

Labeled assumptions, not sourced: if a senior tech spends 50% of their time on a new hire for 3 months, that's about $11,400 of their time; at 25% for 12 months, about $22,800. We're asking maintenance managers for their real split. Per new tech over six months (~250 coached hours): ~$270 to serve today, ~$120 with our local model, against $1,200 of revenue.

THE RISK

All-shift use (8 h/day) makes a flat $200 license $1.19 an hour against our $1.08 cost: about a 9% margin. With the local model, about 60%. So price by coached hours or tiers, and move routine checks to the local model first.

Glasses hardware and adapters

QUESTION	ANSWER
Why not build our own glasses?	Meta, Google, Samsung and Viture + NVIDIA are solving optics, batteries, cameras, radios, distribution and SDKs at a scale we can't match. Our value starts after the camera: procedures, step state, verification, when to speak, records.
What if Meta closes access?	The engine doesn't care where frames come from: each device is a thin adapter over one camera-and-audio interface. Meta is one of several families (Android XR, NVIDIA XR AI glasses, Android headsets).
How do glasses and licenses work?	Glasses are shared site equipment: a site needs enough pairs for the techs coached at the same time, not one per employee. Software is one license per active trainee, and it moves to the next hire. We don't resell hardware.

Adapter status, Oct 20

- **Camera / upload path** — tested: runs live in real time on recorded video (local layers ~50 ms a frame)
- **Meta adapter (toolkit 1.0)** — built, not device-tested
- **Viture / NVIDIA XR AI adapter** — built, not device-tested
- **Android XR** — planned

Appendix A12

Data rights and privacy

QUESTION	DEFAULT ANSWER
Who owns the manuals and footage?	The customer. Praxis gets a limited license to process them to provide the service.
Can Praxis train on proprietary company video?	Only with contractual permission. Improving our shared models needs a separate, explicit opt-in. Without it, a customer's data stays isolated to that customer.
What leaves the site?	The fast layers can run locally; only event keyframes go to a model (about 14% of frames in our evaluation). Keep as little as possible, for as short as possible.
Workers, unions, monitoring rules	Coaching is for the worker; the employer gets records and aggregates. Face and badge blur is next. Each site's workplace, privacy and union rules come first.

IN SHORT

We never assume we own customer video. Shared-model training happens only on data a customer has opted in.

Safety

SITUATION	WHAT PRAXIS DOES
Safety-critical steps (lockout/tagout)	They stay the tech's job. Praxis prompts and records them, and never approves them.
A procedure before anyone uses it	A manager reviews and approves every compiled procedure before a tech sees it, with each card quoted from the manual.
It can't see well enough	It asks for a better view instead of guessing: 77–98% of blurred or blocked checks, never confidently wrong on them.
It isn't sure	Calibrated confidence decides whether to speak; below the bar it stays quiet or asks. Calibration error fell from 0.22 to 0.02 on held-out checks.
Mistake alerts	Our weakest number today: 4 of 17 alerts named a real annotated error. Every pilot measures it before we claim more.

Risks

RISK	WHAT WE DO
Meta or Google ships a general coach	It answers about what's in view. We go deep in maintenance: procedures, step state, records, pilots on site, and we run on their glasses
Viture + NVIDIA win the safety-glasses slot	Run on Helix via NVIDIA's open XR AI library or its app path, rather than fight it
Glasses don't work end to end yet	Adapters are built but untested on a device; testing at a real site is the $2,500 line (appendix A11)
Mistake detection is weak (4 of 17)	Locked procedures, maintenance data and the tech saying "done" all add evidence. Measured in every pilot
Evaluated on cooking, not maintenance	That's why the grant buys ~80 hours of consented maintenance footage first
Depending on model vendors	The verifier is swappable in one config line, and our local student already handles routine checks
Solo founder	The grant tests demand before hiring. First hires come after the first pilot

Why we don't start with consumers

$3.6B

what Autodesk agreed to pay for MaintainX, an app that puts maintenance procedures on a phone.

$99 → $25

what Frontdoor's expert video help for homeowners fell to, per year, from 2023 to 2024.

The same checklist runs on every machine, every month. Home tasks don't repeat, and the platforms give consumer help away free. Consumers come last.

Founder background

LipTrac (2024–26). Wearable real-time visual speech recognition for assistive transcription. Conv3D + bidirectional GRU on 60,000+ video frames, spatiotemporal feature extraction and sequence decoding, evaluated with 12 participants. Provisional patent US 63/812,158.

OCTOPAS (2023–25). End-to-end autonomous oil-spill remediation: drone and marine robotics, computer vision and engineered sorbents. 217.9% better collection performance than conventional methods. Published in Florida Water Resources Journal. Provisional patent US 63/813,616.

Competitive robotics. FTC team captain (2022–26), World Championship qualifier, Control Awards for autonomous software. PID control, Kalman filtering, sensor fusion, OpenCV detection, custom path planning. Built FTC Coach (25,000+ users in 50+ countries) and FTC Elo (3,000+ recurring users).

Startups and growth. Qurios Ventures co-founder and CMO: MRR from under $1k to $50k, a 14,000+ member community, acquired by Crimson Education for $1.3M. Claisen Health co-founder and CMO: 13-person growth team, $200k angel raise. Acely AI (YC W18): $300k attributable revenue. International Research Olympiad: 43 to 89 countries, $50k+ in sponsorships.

Praxis (Oct 2026). Built solo: a tracking engine with trained event and step models, a 48-clip evaluation harness with a stateless baseline, a company backend that indexes manuals and compiles procedures, and a live app with spoken coaching.

Education and awards. Harvard College, Applied Mathematics + Computer Science (expected 2030). Harvard SPARK Fellow; Actuarial Foundation STEM Stars Scholar.

Revenue targets, certifications and robotics

NOW · 2027

Coach new techs

Paid pilots with maintenance teams. With permission, sessions can become labeled training data.

Target: $0.3M · 10 teams

NEXT · 2028

Our own models

Trained on our pilot footage, on the glasses. Cheaper to run and better at maintenance than rented models.

$1.5M · 50 teams · 5×

THEN · 2029

Certifications through Praxis

Partner with NCCER, NIMS and equipment makers so their hands-on tests run on Praxis. Certified techs bring us their next shop.

$4.5M ARR · 150 teams · 3×

LATER · 2030+

Robots: two paths

A Train our own robot models on data customers opted in

B Sell the data and our step checker to robot labs

$13M → $30M ARR, 2030–31

AI LABS ALREADY PAY FOR DATA, AND IT'S GROWING

$2B+ a year

Mercor's run-rate selling expert data to AI labs, doubled in four months

>$100M a year

what robotics companies already spend buying real-world data

$8.6B by 2030

the AI training-data market, growing about 22% a year